

Real-Time PPE Detection and Worker Identification on Construction Sites Using YOLOv8 and Face Recognition

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Abstract—Personal Protective Equipment (PPE) compliance monitoring is a critical challenge in construction site safety. This paper presents an integrated deep learning system combining automated PPE detection using YOLOv8 with real-time face recognition for worker identification. The model targets three classes—helmet, safety vest, and person—derived from the PPE Kit Detection dataset (Kaggle) after class filtering and remapping, yielding 4,904 annotated instances across 3 classes. Three training experiments are conducted: YOLOv8s trained for 30 epochs, YOLOv8s fine-tuned for 20 additional epochs from the best checkpoint, and YOLOv8m trained for 60 epochs with cosine annealing. The fine-tuned YOLOv8s achieves $mAP@50 = 0.857$ with precision 0.896 and recall 0.805, while YOLOv8m reaches $mAP@50 = 0.831$. A key contribution is the coupling of violation detection with real-time face recognition: when a non-compliant worker is detected, the system identifies them by name and transmits the violation alert with a screenshot to the administrator dashboard. A React-based web application with dedicated user and administrator interfaces completes the system. All experiments run on a Tesla T4 GPU using Ultralytics 8.4.35.

Keywords—PPE detection, YOLOv8, face recognition, worker identification, construction site safety, deep learning, real-time monitoring, violation alert.

I. INTRODUCTION

Construction sites are among the most hazardous work environments worldwide. The International Labour Organization (ILO) estimates that construction accounts for approximately 17% of all fatal occupational accidents despite representing a smaller share of the total workforce [1]. A large proportion of these incidents could be prevented through systematic enforcement of Personal Protective Equipment (PPE) regulations, including the mandatory use of safety helmets and high-visibility vests.

Traditional PPE compliance monitoring depends on periodic human inspections, a method inherently limited in coverage and timeliness. Large construction sites operating continuously across multiple zones cannot be realistically monitored through manual surveillance. This has generated growing interest in automated, computer vision-based systems capable of real-time monitoring across multiple camera feeds.

This paper presents an integrated system that advances beyond anonymous violation detection by coupling PPE detection with real-time face recognition. When a worker is detected without required equipment, the system identifies that individual by name and transmits a structured alert—containing the worker's name and a violation screenshot—to

the administrator dashboard, enabling precise and accountable safety enforcement.

The main contributions of this work are: (1) a systematic comparison of three YOLOv8 training configurations for PPE detection; (2) a preprocessing pipeline for class filtering and remapping from a 9-class dataset to three safety-critical classes; (3) integration of real-time face recognition triggered at the moment of violation for worker-level identification; and (4) a full-stack React web application with distinct user and administrator interfaces, where administrators receive real-time named violation alerts.

The remainder of the paper is organized as follows. Section II reviews related work. Section III describes the dataset and preprocessing. Section IV presents the model and training configurations. Section V details the face recognition module. Section VI reports and discusses experimental results. Section VII describes the web application. Section VIII concludes.

II. RELATED WORK

Early approaches to automated PPE detection relied on classical computer vision methods such as Histogram of Oriented Gradients (HOG) combined with Support Vector Machines (SVM). While effective in controlled conditions, these methods proved brittle in real construction environments due to variable lighting, occlusion, and background clutter.

Deep learning significantly advanced detection robustness. Region-based CNNs (R-CNN) and Faster R-CNN [2] improved detection accuracy, but their two-stage pipeline incurs latencies incompatible with real-time deployment. The YOLO family of single-stage detectors, introduced by Redmon et al. [3], addressed this limitation by formulating detection as a direct regression problem. Studies applying YOLOv4 [4] and YOLOv5 to PPE detection report $mAP@50$ scores of 0.75–0.88 depending on dataset and configuration. YOLOv7 [5] further pushed real-time detection performance before the release of YOLOv8.

YOLOv8, released by Ultralytics in 2023 [6], represents the current state of the art in the YOLO lineage. Its redesigned backbone and decoupled detection head—which separates objectness and classification branches—improve convergence stability and detection accuracy over prior

versions. For worker-level accountability, deep metric learning approaches such as FaceNet [7] and ArcFace [8] have achieved near-human face recognition accuracy. The combination of PPE violation detection with real-time face recognition for individual worker identification remains underexplored, which our system addresses.

III. DATASET AND PREPROCESSING

A. Dataset Description

We use the PPE Kit Detection dataset published on Kaggle by Ketaki Chalke [9], containing annotated images of construction workers captured in varied real-world conditions. The original dataset provides bounding box annotations in YOLO format across 9 distinct classes. The dataset is organized into three splits: train, validation, and test, each located under separate image and label directories (/data/images/{split} and /data/labels/{split}).

We focus on three classes of primary safety relevance: helmet (original class 0), safety vest (original class 2), and person (original class 6). These are remapped to class indices 0, 1, and 2 respectively. All label files are reprocessed to retain only annotations belonging to these three selected classes. Images whose corresponding label files contain no selected-class annotations are discarded. Additionally, label files without a corresponding image file are removed to maintain dataset consistency, as a significant number of orphan label files were found during preprocessing (predominantly in the train split).

TABLE I. ANNOTATED INSTANCE COUNTS AFTER PREPROCESSING

Class	Instances	Class ID
Helmet	1,515	0
Safety Vest	1,400	1
Person	1,989	2
Total	4,904	3

B. Data Augmentation

The following augmentations are applied during YOLOv8s baseline training: random horizontal flipping ($p=0.5$), HSV color-space perturbations (hue ± 0.02 , saturation ± 0.7 , value ± 0.4), random rotation ($\pm 10^\circ$), random scale variation ($\pm 50\%$), shear transformation ($\pm 1.5^\circ$), MixUp ($\alpha=0.2$), and copy-paste augmentation ($p=0.3$). For the YOLOv8s fine-tuning experiment, augmentation intensity is reduced (MixUp $\alpha=0.07$, copy-paste $p=0.15$, rotation $\pm 5^\circ$, scale ± 0.4 , shear $\pm 0.7^\circ$) to limit overfitting. For YOLOv8m, default Ultralytics augmentations are used without MixUp or copy-paste, as specified in the training configuration.

IV. MODEL ARCHITECTURE AND TRAINING

A. YOLOv8 Architecture

YOLOv8 employs a Cross Stage Partial (CSP) bottleneck backbone followed by a Path Aggregation Network (PAN) neck for multi-scale feature fusion. Its decoupled detection head separates the objectness and classification predictions into independent branches, improving convergence stability. We evaluate two model scales: YOLOv8s (small, 21.5 MB) and YOLOv8m (medium, 49.7 MB), representing different trade-offs between computational cost and detection capacity.

B. Training Configurations

Three experiments are conducted on an NVIDIA Tesla T4 GPU (14 GB VRAM) using Google Colab, PyTorch 2.10 (CUDA 12.8), and Ultralytics 8.4.35. The dataset YAML defines 3 classes ($nc=3$) and specifies the train/val/test split paths.

Experiment 1 — YOLOv8s baseline: The model is initialized from COCO pretrained weights (yolov8s.pt, 21.5 MB) and trained for 30 epochs with batch size 16, input resolution 640×640 px, $lr=0.01$, and the augmentation strategy described above. The best checkpoint is saved to /content/runs/detect/ppe_detection/weights/best.pt.

Experiment 2 — YOLOv8s fine-tuning: Starting from the best checkpoint of Experiment 1, training continues for 20 additional epochs with $lr=0.0007$, cosine learning rate scheduling ($\cos_lr=True$), warmup of 3 epochs, freeze of the first 10 layers, and reduced augmentation. Patience is set to 10 epochs. Results are saved to /content/runs/detect/ppe_finetune_v2.

Experiment 3 — YOLOv8m: The medium model is initialized from COCO pretrained weights (yolov8m.pt, 49.7 MB) and trained for 60 epochs with $lr=0.01$, cosine annealing, early stopping patience of 20 epochs, and default augmentations (without MixUp or copy-paste). Results are saved to /content/runs/detect/ppe_m.

V. FACE RECOGNITION INTEGRATION

A. Motivation

Standard PPE monitoring systems produce anonymous violation events associated with a camera zone and timestamp, but cannot identify the specific worker involved. This limits corrective action on large multi-zone sites. By coupling PPE detection with face recognition, the system achieves worker-level accountability: non-compliant individuals are identified by name at the moment of violation, enabling targeted interventions, repeat-offender tracking, and individual-level regulatory documentation.

B. Face Recognition Pipeline

The face recognition module is triggered when the YOLOv8 detector produces a person bounding box with no co-located helmet or vest detection within a configured spatial proximity threshold (indicating a PPE violation). The module crops the face region from the violating person bounding box, computes its facial embedding using a deep face encoder, and performs a nearest-neighbor cosine-similarity search against the enrolled worker database. If the similarity exceeds the recognition threshold, the worker is identified by name; otherwise, the event is logged as involving an unidentified person.

C. Violation Alert

Upon confirmation of both a PPE violation and successful worker identification, the system composes and transmits a structured alert to the administrator interface via WebSocket. The alert payload includes: (1) the identified worker's full name; (2) a screenshot of the violation frame with rendered bounding boxes; (3) the specific missing PPE item(s); and (4) the timestamp. Alerts are simultaneously

persisted in the violation log for audit and compliance reporting.

VI. RESULTS AND DISCUSSION

A. Experimental Results

Table II reports the evaluation metrics for the three training configurations on the validation set (final best checkpoint evaluation reported by Ultralytics). Precision, recall, and mAP@50 are extracted from the results_dict produced at the end of each training run.

TABLE II. VALIDATION PERFORMANCE OF TRAINING CONFIGURATIONS

Model	Epochs	mAP50	Precision	Recall
YOLOv8s	30	0.849	0.883	0.796
YOLOv8s (tuned)	20	0.857	0.896	0.805
YOLOv8m	60	0.831	0.865	0.783

The fine-tuned YOLOv8s (Experiment 2) achieves the highest mAP@50 of 0.857, with precision 0.896 and recall 0.805. Notably, YOLOv8m (Experiment 3) achieves lower mAP@50 (0.831) despite its larger capacity, which may be explained by the absence of MixUp and copy-paste augmentation in Experiment 3 and the relatively small size of the filtered dataset. These results highlight that careful fine-tuning of a smaller model can outperform a larger model trained with default settings on a constrained dataset.

B. Inference Speed

Inference speeds reported by Ultralytics at end of training: YOLOv8s fine-tuned: 4.6 ms per image (inference only); YOLOv8m: 12.4 ms per image. During test-set prediction (Experiment 1), inference time is approximately 15.5–16.2 ms per image at shape (1, 3, 640, 640) including pre/post-processing, across 127 test images. All configurations comfortably exceed the real-time threshold of 33 ms (30 fps), confirming practical deployability.

C. Discussion

All three configurations achieve mAP@50 above 0.83, confirming that YOLOv8 is well-suited to PPE detection on construction sites. Precision is consistently higher than recall across all models, indicating a conservative detection behavior—fewer but more confident predictions—which is acceptable in a safety monitoring context where false positives are less critical than missed violations.

The counter-intuitive underperformance of YOLOv8m relative to both YOLOv8s variants, despite its larger capacity, points to an overfitting effect amplified by the absence of MixUp and copy-paste augmentation in Experiment 3. This suggests that on constrained datasets, augmentation strategy has a greater impact on performance than model scale. Future work should explore recall-oriented threshold tuning and per-class metric analysis for each of the three classes (helmet, vest, person).

VII. WEB APPLICATION ARCHITECTURE

A. System Overview

The trained model and face recognition module are deployed as the backend of a full-stack React web application for real-time PPE compliance monitoring. The system

provides two role-based interfaces: a user dashboard for operators and a privileged administrator interface for safety managers. Sensitive identification data, violation records, and configuration capabilities are restricted to administrators.

B. User Interface (React)

The user interface is a React single-page application. It streams camera input to the backend inference server and displays annotated frames with bounding boxes and class labels in real time. A compliance status panel flags detected persons without co-located PPE detections. Users can browse a historical violation log filterable by zone, time range, and type; worker names are shown where face recognition succeeded.

C. Administrator Interface

The administrator interface is the primary destination for face-recognition-coupled violation alerts. When a PPE violation is confirmed and a worker identified, a WebSocket alert is pushed to the administrator screen containing: the worker's full name, a violation screenshot with bounding box overlays, the missing equipment type, the camera zone, and the timestamp. This enables the safety manager to take immediate targeted corrective action.

Additional administrator capabilities include: camera feed management, recognition confidence threshold configuration, worker enrollment management (adding or removing facial embeddings), and compliance reporting with exportable PDF and CSV summaries covering violation trends, zone-level compliance rates, and per-worker incident history.

D. Backend Infrastructure

The RESTful API backend manages camera streams, routes inference requests to the YOLOv8 engine, triggers the face recognition module on violation events, and persists records in a database. WebSocket connections stream annotated frames and named violation alerts to connected clients. The worker enrollment store maintains facial embeddings indexed by name, supporting fast cosine-similarity nearest-neighbor lookup at inference time.

VIII. SOURCE CODE AVAILABILITY

The complete implementation of the proposed system is publicly available on GitHub:

https://github.com/OumaimaKrt/PPE_Detection

IX. CONCLUSION

This paper presented an integrated system for automated PPE detection and worker identification on construction sites, combining YOLOv8 with real-time face recognition and a React-based web application. The dataset preprocessing pipeline filtered a 9-class Kaggle dataset to three safety-critical classes (helmet, vest, person), yielding 4,904 annotated instances. Three training configurations were evaluated: a YOLOv8s baseline (30 epochs, mAP@50 = 0.849), a fine-tuned YOLOv8s (20 additional epochs, mAP@50 = 0.857, precision = 0.896, recall = 0.805), and a YOLOv8m (60 epochs, mAP@50 = 0.831). All models achieve inference speeds well below the 33 ms real-time threshold. The system's key innovation is the coupling of violation detection with face recognition, enabling the administrator to receive a named alert with a violation screenshot immediately upon detection. Future work will

include improvement of face recognition robustness under adverse conditions, per-class metric evaluation, and exploration of edge deployment options.

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